

Lagrange Duality

Primal problem:

$$\min_x f_0(x)$$

$$f_i(x) \leq 0, \quad i = 1, \dots, n$$

$$h_j(x) = 0, \quad j = 1, \dots, m$$

$$\lambda_i$$

$$\nu_j$$

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$$L(x, \lambda, \nu) := f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$$

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Dual objective:

$$g(\lambda, \nu) := \inf_x L(x, \lambda, \nu)$$

Dual problem:

$$\max_{\lambda, \nu} g(\lambda, \nu)$$

$$g(\lambda, \nu) > -\infty$$

$$\lambda \geq 0$$

Lagrange Duality for LPs

Transform to $=0$, ≤ 0 :

$$\min_x \langle c, x \rangle$$

$$Ax = b$$

$$x \geq 0.$$

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$$\begin{array}{ccc} \min_x \langle c, x \rangle & & \min_x \langle c, x \rangle \\ Ax = b & \longrightarrow & Ax - b = 0 \\ x \geq 0. & & -x \leq 0. \end{array} \quad \begin{array}{c} \nu \\ \lambda \end{array}$$

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Dual function:

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \inf_x \left(-\langle \nu, b \rangle + \langle c - \lambda + A^\top \nu, x \rangle \right)$$

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**When does
the minimum
exist?**

$$\inf_x (\dots) > -\infty \quad \Rightarrow \quad c - \lambda + A^\top \nu = 0$$

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Dual function:

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \inf_x (-\langle \nu, b \rangle + \langle c - \lambda + A^\top \nu, x \rangle)$$

**When does
the minimum
exist?**

$$\inf_x (\dots) > -\infty \quad \Rightarrow \quad c - \lambda + A^\top \nu = 0$$

$$\max_{\nu, \lambda} -\langle \nu, b \rangle$$

Dual problem:

$$\begin{aligned} c - \lambda + A^\top \nu &= 0, \\ \lambda &\geq 0 \end{aligned}$$

Lagrange Duality for LPs: Exercise

$$\min -x_2$$

$$x_1 - x_2 - 1 \geq 0$$

$$x_1 + x_2 - 1 \geq 0$$

$$x_1 \leq 2$$

Duality Gap

Theorem (Weak duality): For any $\lambda \geq 0$ and any ν it holds

$$g(\lambda, \nu) \leq f_0(x^*) = p^*$$

Theorem (Strong duality): Let f_i be convex and h_j linear. Then the optimal dual value is equal to optimal primal one.

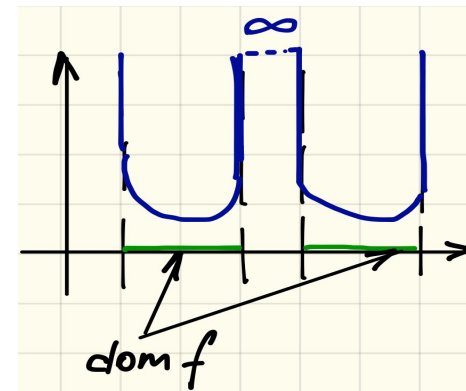
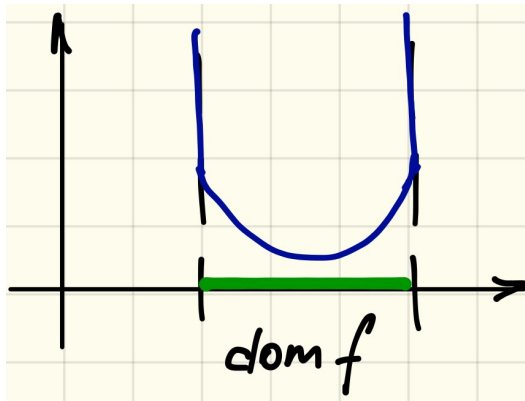
$$\begin{array}{ll} \min_x f_0(x) & = \max_{\lambda, \nu} g(\lambda, \nu) \\ f_i(x) \leq 0, \quad i = 1, \dots, n & g(\lambda, \nu) > -\infty \\ h_j(x) = 0, \quad j = 1, \dots, m & \lambda \geq 0 \end{array}$$

Domain of a Convex Function is Convex

Proposition: f -convex $\Rightarrow \text{dom } f$ -convex

Proof:

$$-\infty < f(x), -\infty < f(y) \Rightarrow -\infty < \alpha f(x) + (1 - \alpha)f(y) \leq f(\alpha x + (1 - \alpha)y)$$

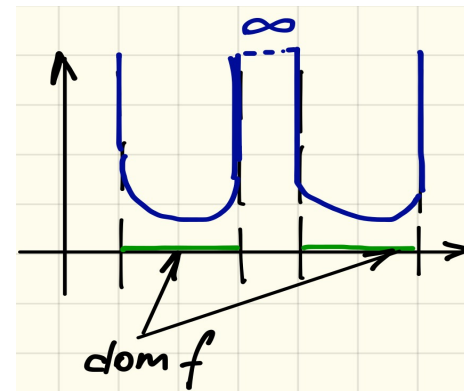
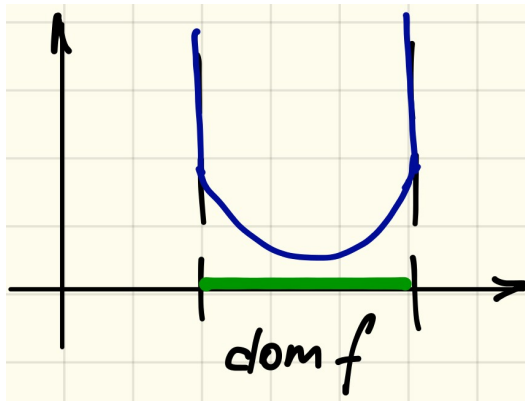


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Proposition: f -concave $\Rightarrow \text{dom } f$ - ?

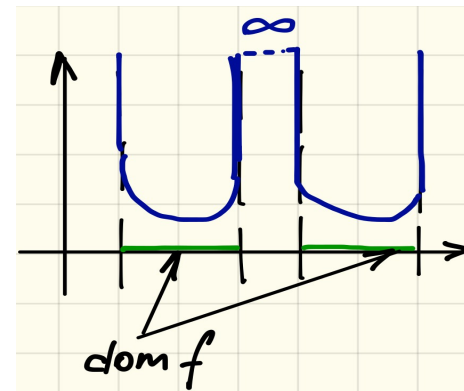
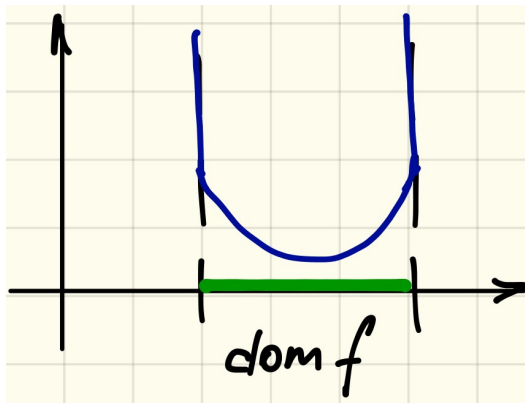
- 1) convex
- 2) non-convex
- 3) I do not know

Domain of a Convex Function is Convex

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$$-\infty < f(x), -\infty < f(y) \Rightarrow -\infty < \alpha f(x) + (1 - \alpha)f(y) \geq f(\alpha x + (1 - \alpha)y)$$



Proposition: f -concave $\Rightarrow \text{dom } f$ -convex

Proof:

$$f(x) > -\infty, f(y) > -\infty$$

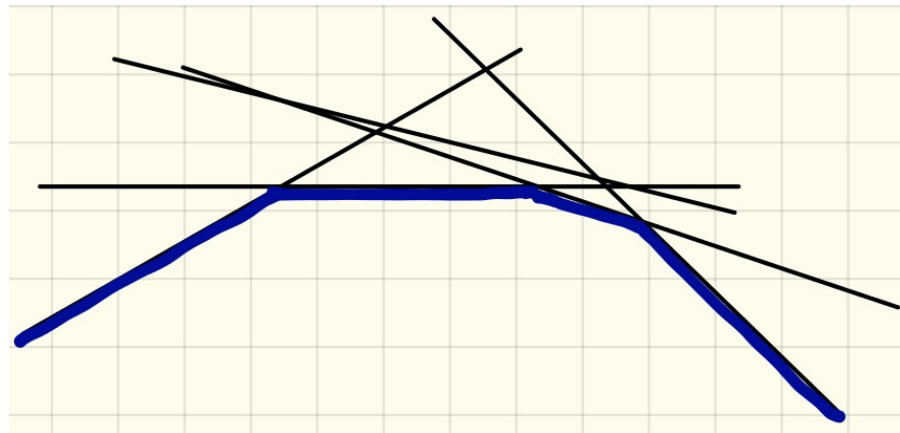
$$\Rightarrow -\infty < f(\alpha x + (1 - \alpha)y) \geq \alpha f(x) + (1 - \alpha)f(y) > -\infty$$

Concavity of the Dual

Proposition: Dual function $g(\lambda, \mu)$ is concave.

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \inf_x f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$$

$L(\cdot, \lambda, \nu)$ – linear on λ, ν

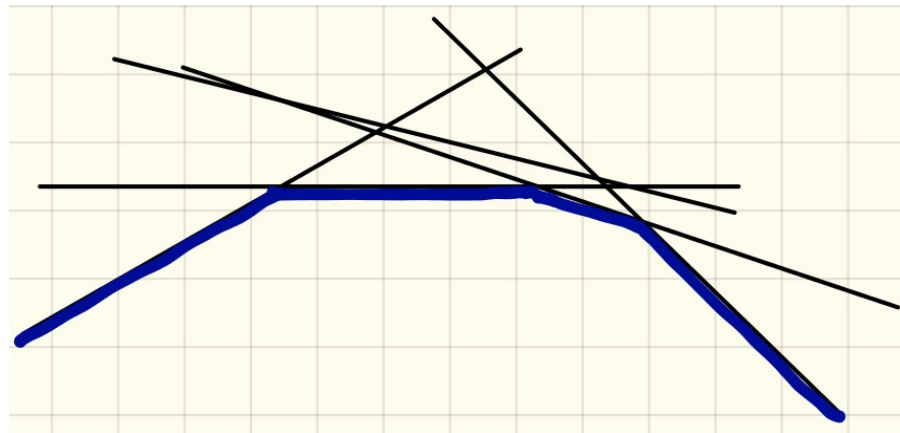


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$L(\cdot, \lambda, \nu)$ – linear on λ, ν



Dual is concave even for a non-convex primal!

Concavity of the Dual

Proposition: Dual function $g(\lambda, \mu)$ is concave.

The dual problem is:

$$\max_{\lambda, \nu} g(\lambda, \nu)$$

$$g(\lambda, \nu) > -\infty$$

$$\lambda \geq 0$$

?

- 1) convex, because g is linear
- 2) concave, because g is a linear function
- 3) concave, because g is concave
- 4) non-convex and non-concave, because of the strict inequality
- 5) non-convex and non-concave, because of the opposite inequality signs
- 6) None of the above

Maximal Lower Bound – Concave Problem

Theorem (Weak duality): For any $\lambda \geq 0$ and any ν it holds

$$\max_{\lambda \geq 0} g(\lambda, \nu) \leq f_0(x^*) = p^*$$

Theorem (Strong duality): Let f_i be convex and h_j linear. Then the optimal dual value is equal to optimal primal one.

$$\begin{array}{ll} \min_x f_0(x) & = \max_{\lambda, \nu} g(\lambda, \nu) \\ f_i(x) \leq 0, \ i = 1, \dots, n & g(\lambda, \nu) > -\infty \\ h_j(x) = 0, \ j = 1, \dots, m & \lambda \geq 0 \end{array}$$

Dual to Maximization Problems

Primal:

$$\max_x f_0(x) \leq$$

$$f_i(x) \leq 0, \quad i = 1, \dots, n \quad \left| \quad \lambda_i$$

$$h_j(x) = 0, \quad j = 1, \dots, m \quad \left| \quad \nu_j$$

Dual:

$$\min_{\lambda, \nu} g(\lambda, \nu)$$

$$g(\lambda, \nu) < \infty$$

$$\lambda \leq 0$$

$$L(x, \lambda, \nu) = \dots$$

$$\inf_x \rightarrow \sup_x$$

$$g(\lambda, \nu) < \infty$$

$$\lambda \geq 0 \rightarrow \lambda \leq 0$$

Example: Dual to Maximization LP

Primal problem:

$$\begin{array}{ll} \max_{\nu} & \langle \nu, b \rangle \\ A^{\top} \nu \leq c & \mid x \end{array}$$

Example: Dual to Maximization LP

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$$\begin{array}{ll} \max_{\nu} & \langle \nu, b \rangle \\ & A^{\top} \nu \leq c \quad | \quad x \end{array}$$

Lagrangian:

$$L(\nu, x) = \langle \nu, b \rangle + \langle x, A^{\top} \nu - c \rangle = \langle \nu, Ax + b \rangle - \langle c, x \rangle$$

Example: Dual to Maximization LP

Primal problem:

$$\begin{array}{ll} \max_{\nu} & \langle \nu, b \rangle \\ & A^{\top} \nu \leq c \quad | \quad x \end{array}$$

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$$L(\nu, x) = \langle \nu, b \rangle + \langle x, A^{\top} \nu - c \rangle = \langle \nu, Ax + b \rangle - \langle c, x \rangle$$

$$\min_x - \langle c, x \rangle$$

$$Ax + b = 0$$

$$x \leq 0$$

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$$\min_x - \langle c, x \rangle$$

$$Ax + b = 0$$

$$x \leq 0$$

$$x \rightarrow -x$$



$$\min_x \langle c, x \rangle$$

$$Ax = b$$

$$x \geq 0$$

Example: Dual to Maximization LP

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$$\begin{array}{ll} \max_{\nu} & \langle \nu, b \rangle \\ & A^{\top} \nu \leq c \quad | \quad x \end{array}$$

Dual problem:

$$\begin{array}{ll} \min_x & \langle c, x \rangle \\ & Ax = b \\ & x \geq 0 \end{array}$$

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$$\min_x - \langle c, x \rangle$$

$$Ax + b = 0$$

$$x \leq 0$$

$$x \rightarrow -x$$



$$\min_x \langle c, x \rangle$$

$$Ax = b$$

$$x \geq 0$$

Example: Dual to Maximization LP

Primal problem:

$$\begin{array}{ll} \max_{\nu} \langle \nu, b \rangle \\ A^{\top} \nu \leq c & | \quad x \end{array}$$

Dual problem:

$$\begin{array}{ll} \min_x \langle c, x \rangle \\ Ax = b \\ x \geq 0 \end{array}$$

Primal problem:

$$\begin{array}{ll} \min_x \langle c, x \rangle \\ Ax = b \\ x \geq 0. \end{array}$$

Dual problem:

$$\begin{array}{ll} \max_{\nu} \langle \nu, b \rangle \\ A^{\top} \nu \leq c \end{array}$$

Dual to dual LP = primal LP (holds for convex problems in general)

Dual to a Maximization Problem

Proposition: Dual function $g(\lambda, \mu)$ to a maximization problem is ?

$$g(\lambda, \nu) = \sup_x L(x, \lambda, \nu) := \sup_x f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$$

- 1) convex, a supremum of linear functions
- 2) concave, as a supremum of linear functions
- 3) linear, as a supremum of linear functions
- 4) None of the above
- 5) I do not know

Dual to a Maximization Problem

Proposition: Dual function $g(\lambda, \mu)$ to a maximization problem is convex

$$g(\lambda, \nu) = \sup_x L(x, \lambda, \nu) := \sup_x f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$$

Dual problem is convex as well:

$$\min_{\lambda, \nu} g(\lambda, \nu)$$

$$g(\lambda, \nu) < \infty$$

$$\lambda \leq 0$$

Keywords:

- primal
- dual
- convex
- concave
- lower/upper bound
- duality gap
- weak/strong duality

What about your own summary?